

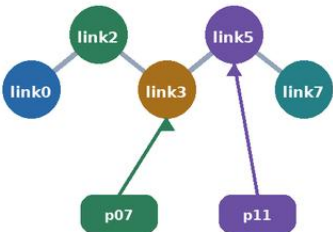
Franka link ownership frames

Each primitive keeps the source link owner before simulation consumption.

LINK GRAPH

Link graph

primitives stay attached to their source Franka link



BODY ATTACHMENT

Attachment table

owner links map to Newton bodies before

prim	owner link	body	allowed
p03	link2	body2	yes
p07	link3	body3	yes
p11	link5	body5	yes
p12	link7	body7	yes

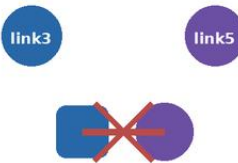


MERGE VIOLATION

Merge violation

cross
struc

owner(p_a) != owner(p_b)



reject cross-link merge

What this shows

The structural reason cross-link primitive merges are rejected.

What this does not show

It is not a whole-robot collision-quality claim.